

1)

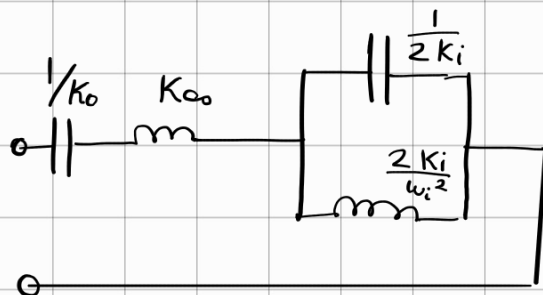
$$F(s) = \frac{s^4 + 4s^2 + 3}{s(s^2 + 2)} \leftarrow \text{NO Dissipative}$$

a) Foster Serie

$$Z(s) = K_0 \frac{1}{s} + K_\infty s + \sum \frac{2K_i s}{s^2 + \omega_i^2}$$

$$\bullet \lim_{s \rightarrow 0} F(s) = K_0 \frac{1}{s} \rightarrow \lim_{s \rightarrow 0} s F(s) = K_0$$

$$\hookrightarrow K_0 = \frac{3}{2} \rightarrow \text{---} \frac{2}{3} \text{---}$$



$$\hookrightarrow Z_1(s) = \frac{s^4 + 4s^2 + 3}{s(s^2 + 2)} - \frac{\frac{3}{2}}{s} = \frac{s^4 + 4s^2 + 3 - \frac{3}{2}s^2 - 3}{s(s^2 + 2)}$$

$$\boxed{Z_1(s) = \frac{s^3 + \frac{5}{2}s}{s^2 + 2}}$$

$$\bullet \lim_{s \rightarrow \infty} Z(s) = K_\infty s^1 \rightarrow \lim_{s \rightarrow \infty} \frac{1}{s} Z(s) = K_\infty$$

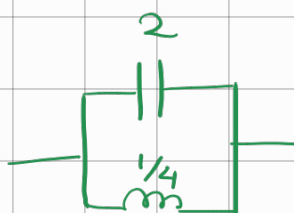
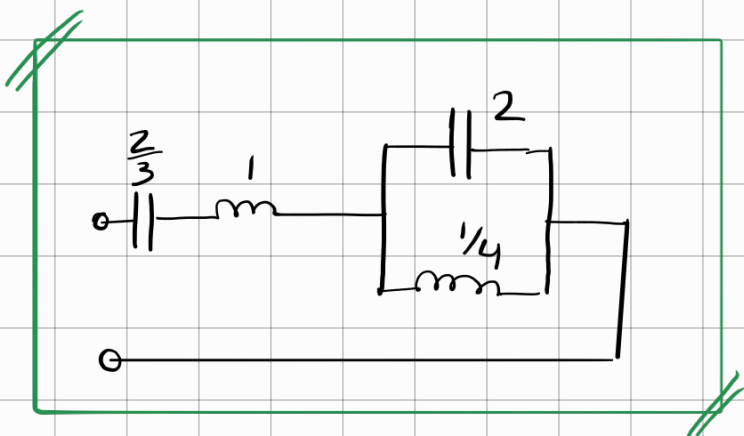
$$\hookrightarrow K_\infty = 1 \rightarrow \text{---} 1 \text{---}$$

$$\hookrightarrow Z_2(s) = \frac{s^3 + \frac{5}{2}s}{s^2 + 2} - s \rightarrow \boxed{Z_2(s) = \frac{\frac{1}{2}s}{s^2 + 2}}$$

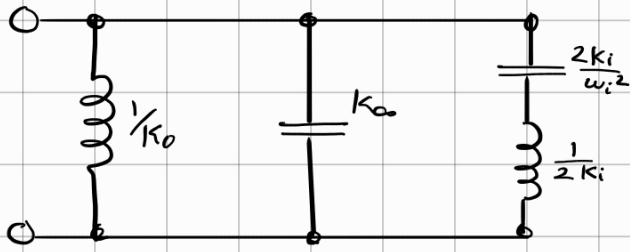
$$\bullet \lim_{s^2 \rightarrow (-2)} Z_2(s) = \frac{2K_i s}{s^2 + 2} \rightarrow \lim_{s^2 \rightarrow (-2)} \frac{s^2 + 2}{2s} \frac{\frac{1}{2}s}{s^2 + 2} = K_i \Rightarrow$$

$$K_i = \frac{1}{4}$$

$$\omega_i^2 = 2$$

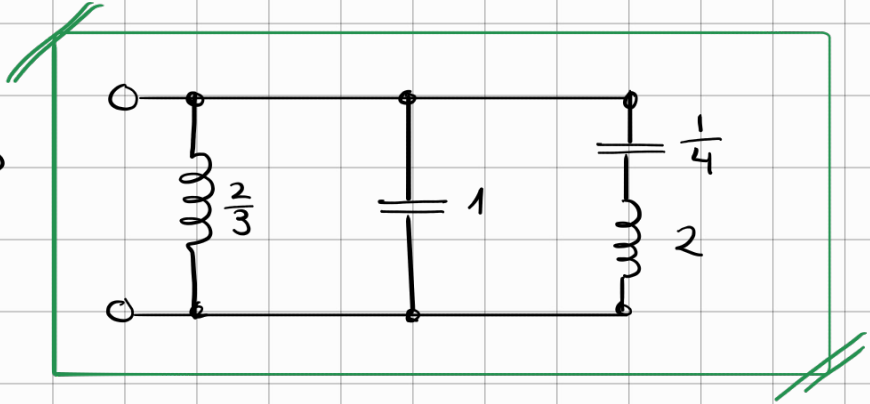


b) Foster Derivación: $Y(s) = k_0 \frac{1}{s} + k_{\infty} s + \sum \frac{2K_i s}{s^2 + \omega_i^2}$



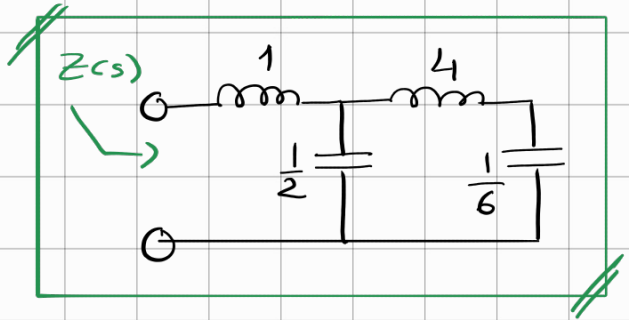
$$F(s) = \frac{s^4 + 4s^2 + 3}{s(s^2 + 2)}$$

$$\begin{cases} K_0 = \frac{3}{2} \\ K_\infty = 1 \\ K_i = \frac{1}{4} \\ \omega_i^2 = 2 \end{cases}$$



c) Caeer en infinito

$$a. F(s) = \frac{s^4 + 4s^2 + 3}{s(s^2 + 2)}$$



$$\begin{array}{r} \rightarrow 5^4 + 4s^2 + 3 \quad | \quad s^3 + 2s \\ - (s^4 + 2s^2) \\ \hline \end{array}$$

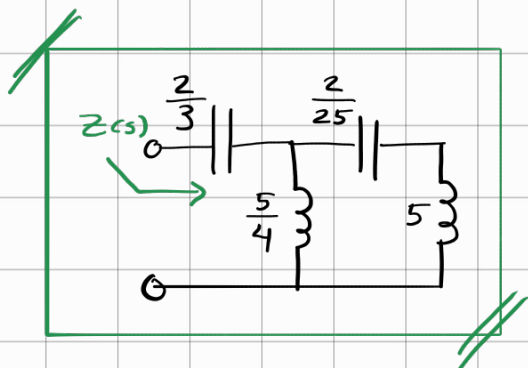
$$\begin{array}{r} 5^3 + 25 \\ -(5^3 + \frac{3}{2}5) \\ \hline \end{array} \quad \begin{array}{r} 25^2 + 3 \\ \hline \frac{1}{2}5 \end{array}$$

$$\begin{array}{r} 2s^2 + 3 \\ - (2s^2) \\ \hline \end{array} \quad \begin{array}{r} \frac{1}{2}s \\ \hline 4s \end{array}$$

$$\begin{array}{r} \frac{1}{2} s \\ - (\frac{1}{2} s) \\ \hline \end{array} \quad \begin{array}{r} 3 \\ \underline{\frac{1}{6} s} \end{array}$$

d) Caver en DC

$$F(s) = \frac{3 + 4s^2 + s^4}{2s + s^3}$$

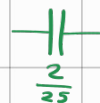


$$\rightarrow \frac{3 + 4s^2 + s^4}{2s + s^3} \left| \frac{2s + s^3}{-(3 + \frac{3}{2}s^2)} \right. \quad \left(\frac{3}{2} \frac{1}{s} \right)$$



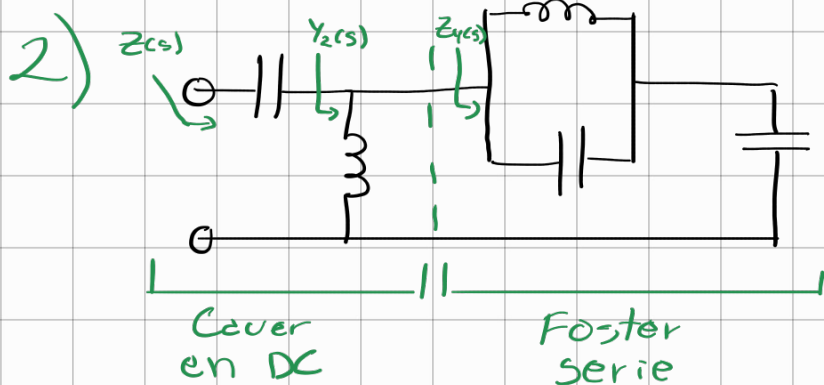
$$\frac{2s + s^3}{-(2s + \frac{4}{5}s^3)} \left| \frac{\frac{5}{2}s^2 + s^4}{(\frac{4}{5} \frac{1}{s})} \right. \quad \left. \frac{5}{4} \right\}$$

$$\frac{\frac{5}{2}s^2 + s^4}{-(\frac{5}{2}s^2)} \left| \frac{\frac{1}{5}s^3}{(\frac{25}{2} \frac{1}{s})} \right. \quad \left(\frac{25}{2} \frac{1}{s} \right)$$



$$\frac{\frac{1}{5}s^3}{-(\frac{1}{5}s^3)} \left| \frac{s^4}{(\frac{1}{5} \frac{1}{s})} \right. \quad \left(\frac{1}{5} \frac{1}{s} \right) \quad \left. \frac{1}{5} \right\}$$

✓



$$Y(s) = \frac{s^5 + 18s^3 + 48s}{6s^4 + 42s^2 + 48}$$

$$Z(s) = \frac{N}{Q} = C + \frac{R}{Q}$$

$$Z(s) = \frac{48 + 42s^2 + 6s^4}{48s + 18s^3 + s^5}$$

← Removemos el capacitor en serie

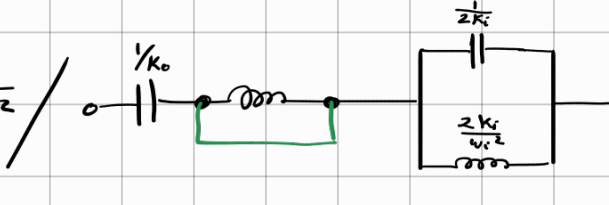
$$Z \rightarrow \frac{48 + 42s^2 + 6s^4}{48s + 18s^3 + s^5} \left| \frac{48s + 18s^3 + s^5}{-(48 + 18s^2 + s^4)} \right. \quad \left(\frac{1}{s} \right)$$



$$Y_2 \rightarrow \frac{48s + 18s^3 + s^5}{48s + 10s^3} \left| \frac{24s^2 + 5s^4}{(2 \frac{1}{s})} \right. \quad \left(2 \frac{1}{s} \right) \quad \left. \frac{1}{2} \right\}$$

$$8s^3 + s^5$$

$$Y_4(s) = \frac{8s^3 + s^5}{24s^2 + 5s^4} = \frac{8s + s^3}{24 + 5s^2} \rightarrow \boxed{Z_4(s) = \frac{5s^2 + 24}{s^3 + 8s}}$$

$$Z_F(s) = K_0 \frac{1}{s} + \cancel{K_\infty s} + \sum \frac{2K_i s}{s^2 + \omega_i^2}$$


$$K_0 \frac{1}{s} = \lim_{s \rightarrow 0} Z_4(s)$$

$$\hookrightarrow K_0 = \lim_{s \rightarrow 0} \frac{5s^2 + 24}{s^2 + 8} \Rightarrow K_0 = 3 \rightarrow \frac{1}{\frac{1}{3}}$$

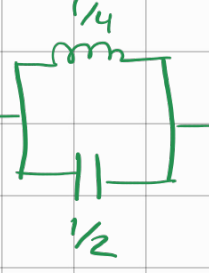
$$\hookrightarrow Z_6(s) = Z_4(s) - K_0 \frac{1}{s} = \frac{5s^2 + 24}{s^3 + 8s} - \frac{3}{s} = \frac{5s^2 + 24 - 3s^2 - 24}{s(s^2 + 8)}$$

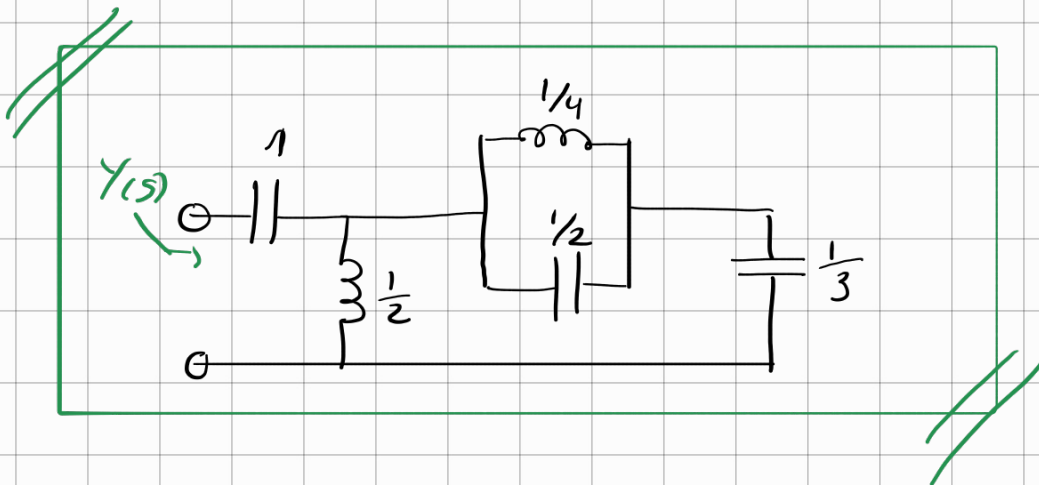
$$\boxed{Z_6(s) = \frac{2s}{s^2 + 8}}$$

$$\frac{5s^2 + 24}{s^2}$$

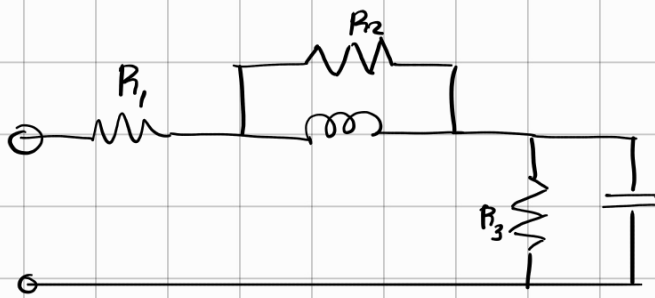
Por inspección:

$$\omega_i^2 = 8$$

$$K_i = 1$$




3)



$$Z(s) = \frac{s^2 + 6s + 8}{s^2 + 6s + 5}$$

$$\circ Z(s) = \frac{(s+2)(s+4)}{(s+1)(s+5)} \quad / \quad \begin{cases} Z(0) = R_1 + R_3 = \frac{8}{5} \\ Z(\infty) = R_1 + R_2 = 1 \end{cases}$$

$$\bullet \text{ Por topología: } Z_{\text{Top}}(s) = R_0' + \frac{k_1' s}{s + \sigma_1'} + \frac{k_2'}{s + \sigma_2'}$$

• Por Fracciones Simples (mismo orden Num y Den)

$$Z_{FS}(s) = R_0 + \frac{k_1}{s + \sigma_1} + \frac{k_2}{s + \sigma_2}$$

$$L, \begin{cases} k_2 > 0 & \text{ya que cumple con la topología} \\ k_1 < 0 & \text{ya que no cumple con la topología} \end{cases}$$

$$\rightarrow \lim_{s \rightarrow -\sigma} Z(s) = \frac{k}{s + \sigma}$$

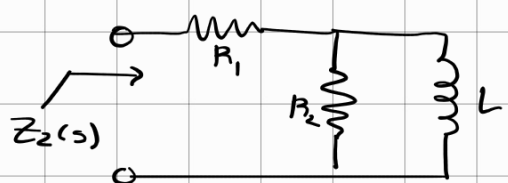
$$\Rightarrow \lim_{s \rightarrow -5} Z(s) = \frac{k}{s + 5} \rightarrow k = \lim_{s \rightarrow -5} \frac{(s+2)(s+4)}{s+1} \rightarrow \boxed{k_1 = -\frac{3}{4}} < 0$$

$$\Rightarrow \lim_{s \rightarrow -1} Z(s) = \frac{k}{s + 1} \rightarrow k = \lim_{s \rightarrow -1} \frac{(s+2)(s+4)}{s+5} \rightarrow \boxed{k_2 = \frac{3}{4}} > 0$$

• $\frac{3/4}{s+1}$ es el término que pertenece al término RC.
 $L, \text{ Se retira.}$

$$Z_2(s) = \frac{s^2 + 6s + 8}{(s+1)(s+5)} - \frac{3/4}{s+1} = \frac{(s+1)(s+17/4)}{(s+1)(s+5)} = \frac{s + 17/4}{s+5} \rightarrow \boxed{Z_2(s) = \frac{s + 17/4}{s+5}}$$

$$L, \begin{cases} Z_2(0) = \frac{17}{20} = R_1 \\ Z_2(\infty) = 1 = R_1 + R_2 \end{cases}$$



$$\Rightarrow \boxed{R_1 = \frac{17}{20}} \quad \wedge \quad \boxed{R_2 = \frac{3}{20}} \quad \wedge \quad \boxed{R_3 = \frac{3}{4}}$$

0,85 0,15 0,75

• Retiramos el ~~mínimo~~ valor de continua

$$\hookrightarrow Z_2'(s) = \frac{s + \frac{17}{4}}{s + 5} - \overset{R_1}{\frac{17}{20}} = \frac{s \frac{3}{20}}{s + 5} \Rightarrow \boxed{Z_2'(s) = \frac{s \frac{3}{20}}{s + 5}}$$

↖ Tanques $RL / Z_{RL} = \frac{sR}{s + R/L}$

$$\hookrightarrow \frac{R}{L} = 5 \quad / \quad R = \frac{3}{20}$$

$$\Rightarrow \begin{cases} R_1 = 17/20, & C = 4/3 \\ R_2 = 3/20, & L = \frac{3}{100} \\ R_3 = 3/4 \end{cases}$$

